V1.2.0

THIRTYSOMETHING

TSTemplate

Configurable template
for LaTeX articles/books

ThirtySomething

11/05/2026

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Change history

Version	Date	Section	Description	Name
1.0.0	30.03.2022	–	• Initial revision	ThirtySomething
1.0.1	30.03.2022	–	• Fix usage of title in header.	ThirtySomething
1.0.2	30.03.2022	–	• Translate last german sentence.	ThirtySomething
1.0.3	01.04.2022	–	• Translate german parts to english. • Introduction of meta data for template file names.	ThirtySomething
1.0.4	02.04.2022	–	• Generic name for meta data file	ThirtySomething
1.0.5	15.04.2022	–	• FIX: Replace TSMeta.tex with TSTemplate-Meta.tex in TSContent.tex • NEW: Added commands for arrows • FIX: All captions in footnotesize	ThirtySomething
1.0.6	20.04.2022	–	• NEW: Added commands/template for appendices	ThirtySomething
1.0.7	28.04.2022	–	• NEW: Added tsFontStrikethrough	ThirtySomething
1.0.8	30.04.2022	–	• NEW: Introduce switches for features in meta file • CHG: Rename TSTitle to TSTitlePage • CHG: Rename commands, use tsText. . . instead of tsFont. . . • NEW: Add subsection naming conventions	ThirtySomething
1.0.9	01.05.2022	–	• NEW: Add template for book also	ThirtySomething
1.0.10	03.05.2022	–	• NEW: Images with overlaid text	ThirtySomething
1.0.11	06.05.2022	–	• NEW: Merge article and book templates, use switch instead • NEW: Rename files to support nested file feature of VSCode	ThirtySomething
1.0.12	03.06.2022	–	• FIX: Use linewidth instead of textwidth	ThirtySomething
1.0.13	16.06.2022	–	• FIX: Use logo on title page only if configured • NEW: Use variable in meta file for content language definition • NEW: Use variable in meta file for page label • NEW: Command for a frac • NEW: Command for a diameter • NEW: Add index and trigger variable • NEW: Add references and trigger variable • NEW: Command for a citation	ThirtySomething
1.0.14	17.06.2022	–	• FIX: Dottet lines in table of content • FIX: Oversized URLs in bibliography with word wrap • FIX: Image with frame and overlay now with correct margins • NEW: Use variable in meta file to influence indentation • UPD: Order commands in internal und public commands • FIX: Numeric footnotes on tables • NEW: Use URL when no logo is used	ThirtySomething
1.0.15	25.06.2022	–	• NEW: For books empty page after title page • UPD: Change order of book parts	ThirtySomething
1.0.16	05.10.2022	–	• NEW: Command tsNP to start a new paragraph	ThirtySomething
1.0.17	02.08.2023	–	• NEW: Split content file into single file per section	ThirtySomething
1.0.18	07.09.2023	–	• NEW: Load custom command file only if enabled	ThirtySomething
1.0.19	10.02.2024	–	• Simplify page label	ThirtySomething
1.0.20	16.03.2024	–	• Use package underscore instead of custom command	ThirtySomething
1.0.21	23.03.2024	–	• Use package nicefrac and replace construct in tsFrac	ThirtySomething
1.0.22	05.11.2026	–	• Minor fixes and improvements	ThirtySomething
1.1.0	05.11.2026	–	• Move commands to own package **tscommands.sty**	ThirtySomething
1.1.1	05.11.2026	–	• Additional fixes and improvements	ThirtySomething
1.1.2	05.11.2026	–	• Template and typography improvements	ThirtySomething
1.2.0	05.11.2026	–	• Drop usage of obsolete background package	ThirtySomething

Table 1: Change history

1 License information

This template is subject to the terms of the MIT license:

```
MIT License

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LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.
```

Figure 1: MIT License

2 Introduction

This is a template for use with LaTeX. An attempt has been made to make the usage of this template as simple as possible for the user.

2.1 Naming conventions

For this document some naming conventions are used.

- Text in `monospace` is either a command or LaTeX source code.
- Text in *italic* are usually filenames.

3 Template

The template is divided into several files, which are listed here below:

File	Meaning
LICENSE	MIT license file.
README.md	Information about the template.
TSTemplate.pdf	The result as a PDF file.
TSTemplate.tex	The envelope document.
TSTemplate-Appendix.tex	The appendices of the document.
TSTemplate-Commands.tex	Template-specific layout helpers.
tscommands.sty	Reusable beginner-friendly commands.
TSTemplate-Content.tex	The content of the document.
TSTemplate-CustomCommands.tex	The document specific commands.
TSTemplate-Glossary.tex	The glossary of the document.
TSTemplate-History.tex	Document history.
TSTemplate-Hyphenation.tex	Document-specific hyphenation.
TSTemplate-Logo.png	The logo used.
TSTemplate-Meta.tex	The metadata of the document.
TSTemplate-TitlePage.tex	The title page of the document.

Table 2: The files of the template

Some commands and formatting have been predefined for the template. They all have in common that they start with a `\ts...`. This serves to distinguish them from the commands otherwise used in LaTeX. The available commands are explained in the Commands chapter. They are provided by `tscommands.sty`, which can also be loaded in other documents with `\usepackage{tscommands}`. The template-specific layout helpers remain in *TSTemplate-Commands.tex*.

NOTE: The file containing the meta data is depending on the name of the envelope document *TSTemplate.tex*. The rule is just simple: The name

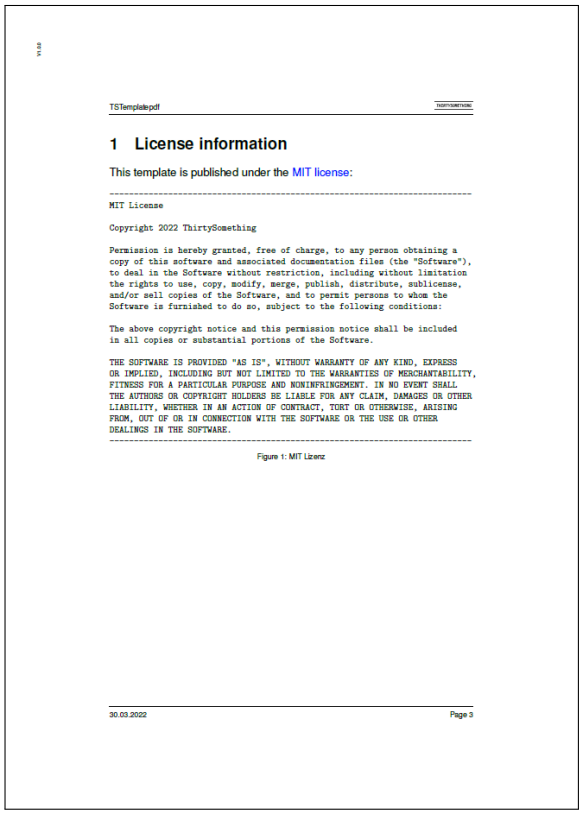
will be *TSTemplate* followed by *-Meta.tex*. If you name your document *MyDocument.tex* then the filename of the metadata file will be *MyDocument-Meta.tex*.

4 Usage

As already described, the use of the template should be simple. So the user copies the template, that is, the entire directory with all files and subdirectories.

Normally the user adjusts the metadata (file *TSTemplate-Meta.tex*, see Metadata I and Metadata II) and the content in the file *TSTemplate-Content.tex* according to his needs. Everything else is managed by the template.

It is recommended to rename the main template file *TSTemplate.tex* into a more more descriptive name.



V1.0.0

TSTemplatepdf THIRTYSOMETHING

1 License information

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Figure 1: MIT Lizenz

30.03.2022 Page 3

Figure 2: Document page

4.1 Metadata I

For each document there are some metadata. This means the following definitions:

Metadefinition	Meaning
\tsAuthor{}	The author of the document.
\tsSubject{}	The subject of the document.
\tsTitle{}	The title of the document.
\tsURL{}	The url of the author.
\tsVersionMajor{}	The major version of the document.[1]
\tsVersionMinor{}	The minor version of the document.[1]
\tsVersionPatch{}	The patch number of the document.[1]
\tsVersionString{}	Complete version string.[1]

Table 3: Metadata I

These metadata are referenced in various places in the template. For example, they are used in the title page of the document. The version number is based on semantic versioning[1].

[1] See Semantic Versioning for more details.

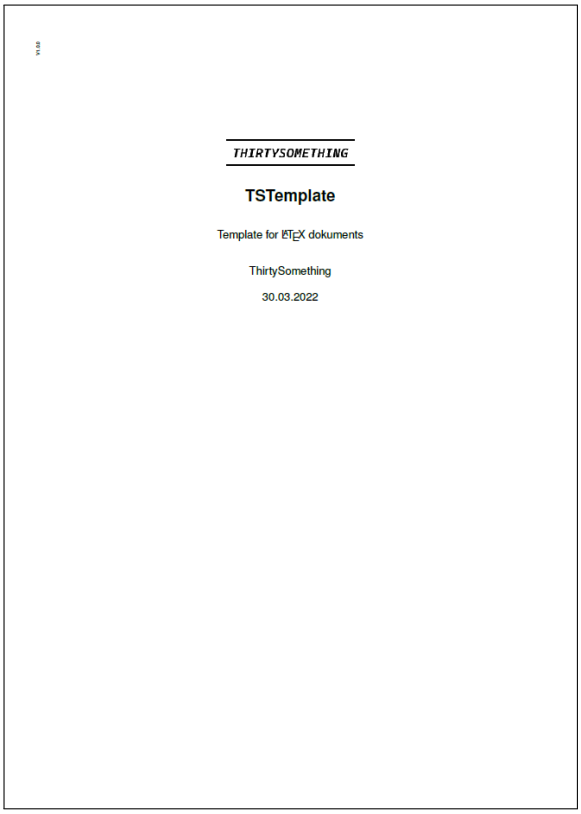


Figure 3: Title page

They are also used to set metadata in the generated PDF:

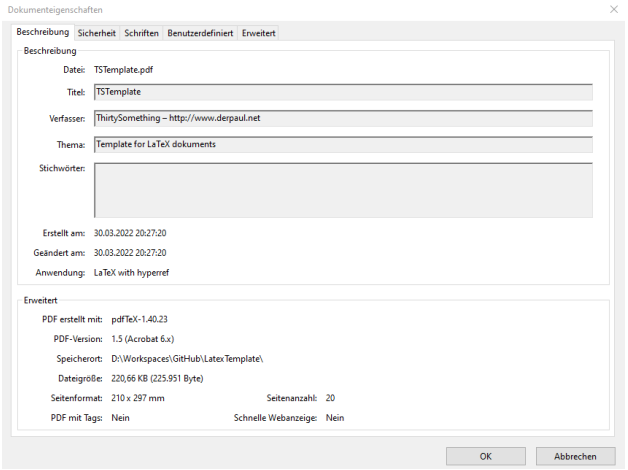


Figure 4: PDF properties

4.2 Metadata II

For further control, there are variables that can be used to enable/disable which features are active for the document. These variables can be found in the file *TSTemplate-Meta.tex*. There are:
Use the corresponding `false` switch to disable a feature. Disabling the glossary skips its package, definitions, and output; guard live glossary commands with `\ifUseTsGlossary`. Disabling citations skips the bibliography, and disabling appendices skips their file; guard live citations and appendix references with their respective switches. Index entries are likewise conditional on `\ifUseTsIndex`. The supplied verbatim examples remain visible regardless of these switches. TOC, figure-list, and table-list switches control list output only, not their underlying content.

Metadefinition	**Beschreibung**

`% Document type: article or book`
`\newcommand{\tsDocumentType}{article}`
`% \newcommand{\tsDocumentType}{book}`
	Toggle between one page layout (article) and two page layout (book)

`% Definition of document language`
`\newcommand{\tsDocumentLanguage}{american}`
`% \newcommand{\tsDocumentLanguage}{ngerman}`
	Setup of used language of document

`% Switch to enable appendix`
`\newif\ifUseTsAppendix`
`% \UseTsAppendixtrue % Disable appendix`
`\UseTsAppendixtrue % Enable appendix`
	Toggle appendix

`% Switch to enable custom commands`
`\newif\ifUseTsCustomCommands`
`% \UseTsCustomCommandstrue % Disable custom commands`
`\UseTsCustomCommandstrue % Enable custom commands`
	Toggle custom commands

`% Switch to enable bibliography and cites`
`\newif\ifUseTsCitation`
`% \UseTsCitationtrue % Disable references`
`\UseTsCitationtrue % Enable references`
	Toggle references

`% Switch to enable glossary`
`\newif\ifUseTsGlossary`
`% \UseTsGlossarytrue % Disable glossary`
`\UseTsGlossarytrue % Enable glossary`
	Toggle glossary

`% Switch to enable index`
`\newif\ifUseTsIndex`
`% \UseTsIndextrue % Disable index`
`\UseTsIndextrue % Enable index`
	Toggle index

Metadefinition	Beschreibung

`% Switch to enable history`
`\newif\ifUseTsHistory`
`% \UseTsHistorytrue % Disable history`
`\UseTsHistorytrue % Enable history`
	Toggle history

`% Switch to enable hyphenation`
`\newif\ifUseTsHyphenation`
`% \UseTsHyphenationtrue % Disable hyphenation`
`\UseTsHyphenationtrue % Enable hyphenation`
	Toggle hyphenation

`% Switch to enable list of figures`
`\newif\ifUseTsLOF`
`% \UseTsLOFtrue % Disable list of figures`
`\UseTsLOFtrue % Enable list of figures`
	Toggle list of figures

`% Switch to enable logo`
`\newif\ifUseTsLogo`
`% \UseTsLogotrue % Disable logo`
`\UseTsLogotrue % Enable logo`
	Toggle logo

`% Switch to enable list of tables`
`\newif\ifUseTsLOT`
`% \UseTsLOTtrue % Disable list of tables`
`\UseTsLOTtrue % Enable list of tables`
	Toggle list of tables

`% Switch to disable indentation of paragraphs`
`\newif\ifUseTsParagraphNoIndentation`
`% Enable indentation:`
`% \UseTsParagraphNoIndentationfalse`
`% Disable indentation:`
`\UseTsParagraphNoIndentationtrue`
	Toggle indentation of paragraphs

`% Switch to enable start of section on new page`
`\newif\ifUseTsSectionOnNewPage`
`% \UseTsSectionOnNewPagetrue % Section on same page`
`\UseTsSectionOnNewPagetrue % Section on new page`
	Toggle start of section

`% Switch to enable title page`
`\newif\ifUseTsTitlePage`
`% \UseTsTitlePagetrue % Disable title page`
`\UseTsTitlePagetrue % Enable title page`
	Toggle title page

`% Switch to enable table of contents`
`\newif\ifUseTsTOC`
`% \UseTsTOCtrue % Disable table of contents`
`\UseTsTOCtrue % Enable table of contents`
	Toggle table of contents

Table 4: Metadata II

It should be noted that only one line has an influence on the option. The last two lines in the option list show the disabled (with the % sign at the beginning of the line) and the enabled (without the % sign) state.

ATTENTION: Turning options off and/or on will affect the steps required for compilation. For example, if the glossary is disabled, no `makeglossary` may be used during compilation. Otherwise this will result in errors.

4.3 Metadata III

Additional to the metadata of section 4.1 there exists metadata to influence the filenames. Except for the *TSTemplate-Meta.tex* file all other files are used indirect by commands. So there are these commands available:

Metadefinition	Meaning
\tsAppendixFile{}	Filename for appendices, default *TSTemplate-Appendix.tex*
\tsCommandsFile{}	Filename for template layout helpers, default *TSTemplate-Commands.tex*
\tsContentFile{}	Filename for content, default *TSTemplate-Content.tex*
\tsCustomCommandsFile{}	Filename for custom commands, default *TSTemplate-CustomCommands.tex*
\tsGlossaryFile{}	Filename for glossary, default *TSTemplate-Glossary.tex*
\tsHistoryFile{}	Filename for history, default *TSTemplate-History.tex*
\tsHyphenationFile{}	Filename for hyphenation, default *TSTemplate-Hyphenation.tex*
\tsLogoFile{}	Filename for logo, default *Images/TSTemplate-Logo.png*
\tsReferencesFile{}	Filename for references, default *TSTemplate.bib*
\tsTitlePageFile{}	Filename for title page, default *TSTemplate-TitlePage.tex*

Table 5: Metadata III

4.4 Document content

The document content is entered like a normal LaTeX document. For this the file *TSTemplate-Content.tex* is used, in which the user enters the text. The basic conditions such as the inclusion of packages are regulated in the file *TSTemplate.tex*. Features are enabled/disabled in the file *TSTemplate-Meta.tex* – see also Metadata II.

4.5 Glossary

The user has the option of entering his own glossary. The entries for the glossary are done in the file *TSTemplate-Glossary.tex*. ThirtySomething is entered here as an example. This looks like this:

```
\newglossaryentry{TS}{name={ThirtySomething},description={A glossary entry}}
```

Figure 5: Definition of a glossary entry

The values have the following correspondences:

- **TS** – This is the abbreviation used to reference the keyword in the document.
- **ThirtySomething** – This is the keyword listed in the glossary.
- **A glossary entry** – This is the description for the keyword listed in the glossary.

The glossary then lists the keyword, the description and the page number(s) where the keyword was used everywhere. A prerequisite for this is, of course, the use of the command `\gls{abbreviation}` with the existing abbreviation. In the above example, the usage looks like this:

```
\gls{TS}
```

Figure 6: Use of a glossary entry

In case a glossary is to be used, the document must be translated with `makeglossaries`.

```
pdflatex
makeglossaries
pdflatex
pdflatex
```

Figure 7: Translate with glossary

4.6 Hyphenation

In some circumstances, hyphenation is not performed for some words. This may be because these words are very specific and LaTeX does not

know them and therefore cannot hyphenate them. The result may be overlong lines, which are clearly visible in most cases. In some cases this overlength is not immediately obvious. For this you can temporarily add the line

```
% To see the (page)frames used for hyphenation determination
% \usepackage{showframe}
```

Figure 8: Preamble – Package showframe

by removing the comment sign (%). This will display frames in the document indicating the text areas – and you can clearly see the overlong lines. The affected words at the end of the line can then be entered in the *TSTemplate-Hyphenation.tex* file. An entry there follows the following pattern:

```
\hyphenation{Thir-ty-Some-thing}
```

Figure 9: Exemplary content of TSHyphenation.tex

An entry corresponds to the word to be separated with a dash after each syllable. If necessary in the different cases, then the words are separated by a space. In addition, the following parameter can be changed in the document *TSTemplate.tex*:

```
% Setting tolerance for word wrap
% See also: https://texfaq.org/FAQ-overfull
\tolerance=3000
```

Figure 10: Influencing the hyphenation

The permissible values are in the range from 0 to 10000.

4.7 Bibliography references

In case you want to have some bibliography references and want to cite, you have to enable the citations in *TSTemplate-Meta.tex*:

```
% Switch to enable bibliography and cites
\newif\ifUseTsCitation
\UseTsCitationtrue
```

Figure 11: Enabled citations

Fill up your bibliography reference file:

```
@misc{lfc,
    title        = {{Latex Font Catalogue}},
    author       = {},
    year         = {},
    note         = {Online checked on 16.06.2022},
    howpublished = {\url{https://tug.org/FontCatalogue/}}
}
```

Figure 12: Reference definition

And to use it in the text.

```
Fonts can be viewed at Latex Font Catalogue.\tsCitation{lfc}
```

Figure 13: Reference usage

Fonts can be viewed at Latex Font Catalogue. [1]

5 Commands

To simplify things for newcomers, various commands have been defined. The use of the commands is optional. They are provided by the `tscommands` package independently of this template. Commands without input take no argument. An empty group after such a command is optional and can delimit its name; it is not consumed as a parameter. Caption, image, and appendix helpers also accept a leading optional reference identifier, independent of the visible text. For example, `\tsCaptionLabelFigure[logo]{Company logo}` creates the label `fig:logo`. All four image helpers accept the same leading optional identifier. The formula helper accepts an optional identifier too, but creates an equation from math source rather than a caption. In the overview below, `[id]` marks this optional argument; replace `id` with your reference identifier or omit it. Without it, the existing caption-based or filename-based label is retained. Use unique identifiers without their prefix, preferably letters, numbers, and hyphens, to keep references stable when captions change. However, using them makes it clearer what the author wants to achieve with the corresponding command. It is recommended to end commands without parameters with {}, even if this is not necessary. However, it leads to the fact that, for example, after such a command there can also be a space character – otherwise this is removed by LaTeX when compiling.

Command definition	**Meaning**
`\tsAppendixIncludeFile[id]{Filename}`	This command will add an appendix section. The section consists of a `tsTextMonospace` section header of `Filename` and will include the file `Filename` in a verbatim environment. With an optional ID, the label is `appendix:id`. See appendix A how it looks like.
`\tsAppendixSection[id]{Appendix}`	This command will add an appendix, also a so-called label. Thus the optional ID creates `appendix:id`; without it, the appendix can be referenced via `\ref{appendix:Appendix}` or via `\nameref{appendix:Appendix}` somewhere else.
`\tsArrowRight{}`	→

Command definition	Meaning
\tsArrowRightDouble{}	⇒
\tsArrowDown{}	↓
\tsArrowDownDouble{}	⇓
\tsArrowLeft{}	←
\tsArrowLeftDouble{}	⇐
\tsArrowUp{}	↑
\tsArrowUpDouble{}	⇑
\tsBackslash{}	\
\tsBullet{}	•
\tsCaptionLabelFigure[id]{Figure}	This command is usually used below a mapping. There is not only a description added, but also a so-called label. The optional ID creates fig:id. Without it, the figure can be referenced via \ref{fig:Figure} or via \nameref{fig:Figure} somewhere else.
\tsCaptionLabelFormula[id] {Formula}	Creates a numbered equation from math source, not a caption. Call it outside math environments, without dollar signs. The optional ID creates for:id, referenced with \ref{for:id}. Without an ID, the label uses the equation number, such as for:1.
\tsCaptionLabelTable[id]{Table}	This command is similar to \tsCaptionLabelFigure. The optional ID creates tab:id. Without it, it is referenced via \ref{tab:Table} or via \nameref{tab:Table}.
\tsCitation{ref}	References citation in bibliography, see also Bibliography references for details.
\tsDegree{zahl}	Without number ° or with number 715°. Works in text and math mode; the argument keeps the surrounding mode.

Command definition	Meaning
`\tsDiameter{}`	⌀
`\tsFootnoteDef{text}{ref}`	Defines a footnote[2] with a label, which can be referenced elsewhere using `\tsFootnoteRef{ref}`.
`\tsFrac{num1}{num2}`	Defines a frac like ½.
`\tsFootnoteRef{ref}`	References a footnote[2] which was defined elsewhere using `\tsFootnoteDef{text}{ref}`.

`\tsImage[id][placement]`
`{filename}{description}`
`{options}`
	With this, an image can be included – it then automatically gets a description. It is mandatory to specify three parameter brackets. The optional placement defaults to `H` (exactly here). Use `htbp` to allow floating; use an empty ID `[][htbp]` when only specifying placement. The first parameter is the file name. The second parameter is the signature under the image. The third parameter, the options, are optional, that is, the brackets can be empty. These are the same parameters that can be used with `\includegraphics`, e.g. `width=380px`.

`\tsImageF[id][placement]`
`{filename}{description}`
`{options}`
	Like `\tsImage`, but with a frame.

`\tsImageO[id][placement]`
`{filename}{description}`
`{overlay}{options}`
	Like `\tsImage`, but with an overlaid text.

`\tsImageOF[id][placement]`
`{filename}{description}`
`{overlay}{options}`
	Like `\tsImageF`, but with an overlaid text.

`\begin{tsLTItemize}`
`  \item Item I of enumeration list.`
`  \item Item B of enumeration list.`
`  \item Item 3 of enumeration list.`
`\end{tsLTItemize}`
	An enumeration list
	• Item I of enumeration list.
	• Item B of enumeration list.
	• Item 3 of enumeration list.

[2] I am a footnote.

Command definition	Meaning
\tsNP{}	Ends the current paragraph and requests extra vertical separation using \bigbreak. For an ordinary paragraph, use a blank source line or \par.
\tsTextBold{text}	**text**
\tsTextItalic{text}	*text*. Selects italic text even inside an italic passage; unlike \emph, it does not toggle emphasis.
\tsTextLower{text}	text
\tsTextMonospace{text}	text
\tsTextQuoteF{text}	≪text≫
\tsTextQuoteG{text}	„text“
\tsTextStrikethrough{text}	~~text~~
\tsTextUnderline{text}	text
\tsTextUpper{text}	text

Table 6: Available commands

For a numbered formula, write \tsCaptionLabelFormula[energy]{E = mc^2}. The result is

$$E = mc^2 \tag{1}$$

and can be referenced as equation 1. A look into the source code of this document shows the usage of the commands.

5.1 Examples to images

```
% Insert image TSTemplate-Logo.png
% Signature is 'Image without parameters'
% Optional parameters are empty
\tsImage{Images/TSTemplate-Logo.png}{Image without parameters}{}
```

Figure 14: Code of image without parameters



Figure 15: Image without parameters

```
% Insert image TSTemplate-Logo.png
% Signature is 'Image with parameter'
% Scale the image to half the local line width:
\tsImage{Images/TSTemplate-Logo.png}
    {Image with parameter}{width=0.5\linewidth}
```

Figure 16: Code of image with parameter



Figure 17: Image with parameter

```
% Insert image TSTemplate-Logo.png with frame
% Signature is 'Image without parameters, with frame'
% Optional parameters are empty
\tsImageF{Images/TSTemplate-Logo.png}
    {Image without parameters, with frame}{}
```

Figure 18: Code of image without parameters, with frame



Figure 19: Image without parameters, with frame

```
% Insert image Garden.jpg with frame
% Signature is 'Image with overlay and parameters'
% Optional parameters are 'width=0.9\linewidth
\tsImageOF{./Images/Garden.jpg}
    {Image with frame, overlay and parameters}
    {\copyright{} 2008, ThirtySomething}{width=0.9\linewidth}
```

Figure 20: Code image with frame, overlay and parameters



Figure 21: Image with frame, overlay and parameters

6 Font sizes

There may be a need to adjust the font size. For this purpose LaTeX provides the following options:
Font-size commands are declarations, not commands with a text argument. Put the declaration and its text inside one group, for example {\Huge Huge}, to keep the size change local. For a whole paragraph, end the paragraph inside the group: {\Huge A paragraph.\par}. This also applies the selected size's line spacing to the paragraph.

Command	**Example**
{\Huge Huge}	Huge
{\huge huge}	huge
{\LARGE LARGE}	LARGE
{\Large Large}	Large
{\normalsize normalsize} (Default)	normalsize (Default)
{\small small}	small
{\footnotesize footnotesize}	footnotesize
{\scriptsize scriptsize}	scriptsize
{\tiny tiny}	tiny

Table 7: Font sizes

7 Tables

Font sizes play a role in the use of tables when it comes to distributing the content appropriately among the columns. Here is a sample code how to make a table.

```
\begin{small}
    \renewcommand*{\arraystretch}{3}
    \begin{longtable}{ | p{0.43\linewidth} | p{0.5\linewidth} | }
        \hline
        \tsTextBold{Command}       & \tsTextBold{Sample} \\
        \hline
        \endhead
        \{\tsBackslash{}Huge text\} & {\Huge Huge}       \\
        \hline
        \tsCaptionLabelTable{Font sizes}
    \end{longtable}
\end{small}
```

Figure 22: Tables

It is started with a block in one of the desired Font sizes. The first command within the font environment changes the spacing between the text lines. If you do not do this, the table content and the table grid will overlap or the spacing will be very tight. A good value is

```
\renewcommand*{\arraystretch}{1.5}
```

Figure 23: Row height in tables

The definition of the columns and their widths follows. Use `\linewidth` for the locally available width. For a table with paragraph columns, its total width includes the sum of the column widths, twice `\tabcolsep` per column, and `\arrayrulewidth` per single vertical rule. This total must not exceed the available width.
For example, a two-column table with three single vertical rules has a column-width budget calculated as follows:

```
\linewidth - 4\tabcolsep - 3\arrayrulewidth
```

This calculation assumes ordinary column padding; custom intercolumn material or double rules require their own allowance. A percentage such as `0.9\linewidth` is only a rule of thumb.

To slightly separate column headings from the column contents, the `\tsTextBold{column name}` command can be used. Each column value is separated from the next column by a `&`. The end of the row is marked with a `\\`.

For horizontal lines, a `\hline` is inserted. Place `\endhead` after the column-heading row and its horizontal rule to repeat those headings on every page. When the first page needs a different heading, define it separately with `\endfirsthead`, followed by the continuation heading ending with `\endhead`.

The data lines follow the same structure.

To give the table a referencable name, add a `\tsCaptionLabelTable{Font sizes}` at the end. Check the source of this document in the section Font sizes to see how it works.

For verbatim source in a table, use a regular `p` column. Select `BVerbatim[baseline=t]` for the code. FancyVerb's top-aligned box aligns the first code line with adjacent text without negative vertical spacing. The template retains `P` as an alias for `p` for compatibility, not as a special alignment mechanism. For a `tsLTItemize` list, use a regular `p` column. The list is top-aligned with the text in adjacent cells and stays together within its table row. Also important is that the column separator `&` must be on the next line after the `\end{tsLTItemize}`.

8 The verbatim environment

With the Verbatim environment text can be displayed in a `Monospace` font. This is used to display source code.

```
\begin{figure}[H]
    \small
    \centering
    \begin{BVerbatim}
        Sample code of verbatim environment.
    \end{BVerbatim}
    \tsCaptionLabelFigure{Verbatim Environment Demo}
\end{figure}
```

Figure 24: Verbatim Environment Code

In order for the verbatim environment to get a signature, it is embedded in a figure environment. The `[H]` behind it makes sure that the environment stays at the desired place in the text and does not end up somewhere in the document.

The adjustment of the font size is done as needed. A possible centering is achieved here using `\centering`.

Note: If the verbatim environment is to be centered, the text that is between `\begin{BVerbatim}` and `\end{BVerbatim}` must be leftmost aligned in the source code of the document. Otherwise the spaces will be part of the alignment.

The example from above then produces this result:

```
Sample code of verbatim environment.
```

Figure 25: Verbatim Environment Demo

9 Appendices

In case you want to have appendices, there are two steps. The first one is required to enable the appendices, the second one is the appendix itself. For enabling remove the comment sign in the *TSTemplate-Meta.tex* file:

```
% Switch to enable appendix
\newif\ifUseTsAppendix
\UseTsAppendixtrue
```

Figure 26: Enabling appendices

The second one is the usage of an appendix. You can have a look at the sources in *TSTemplate-Appendix.tex* or use this example:

```
\tsAppendixSection{README.md}
```

Figure 27: Usage of tsAppendixSection

In this example the appendix `README.md` is added and can be referenced either by `\nameref{appendix:README.md}` (README.md) and/or `\ref{appendix:README.md}` (A).

In case you want to add a file to the appendix, you can also use the `\tsAppendixIncludeFile{filename}` command. You can see the result in appendix C. The source is

```
\tsAppendixIncludeFile{vscode-latex-workshop.json}
```

Figure 28: Usage of tsAppendixIncludeFile

An optional identifier keeps the reference independent of the filename. Use `\tsAppendixIncludeFile[readme]{README.md}`. This creates `appendix:readme`. The same optional identifier is supported by `\tsAppendixSection`.

A README.md

```
# LatexTemplate

A basic template for latex documents. See the [PDF][pdf] for more details.

## Engine and format requirements

The template and ‘tscommands‘ require LaTeX2e dated 2021-06-01 or newer and
support pdfLaTeX, LuaLaTeX, and XeLaTeX. Install the dependencies listed below,
along with BibTeX and makeindex for the default bibliography, index, and
glossary features. The default document has been built with all three engines
on the available TeX Live installation; the minimum kernel date is a required
API floor, not a claim that every historical distribution was tested.

Run one of these commands from the template directory:

‘‘‘sh
latexmk -pdf TSTemplate.tex
latexmk -lualatex TSTemplate.tex
latexmk -xelatex TSTemplate.tex
‘‘‘

Use a clean output directory when switching engines. Hyperref and the graphics
packages select their drivers automatically. The template explicitly retains
OT1 encoding and its existing Type 1 font setup across engines; choosing a
Unicode engine does not automatically switch to system
fonts or ‘fontspec‘.

The sample ‘vscode-latex-workshop.json‘ provides corresponding latexmk recipes.
Its settings must be incorporated into your VS Code configuration to use them.

## Reusable commands

[tscommands.sty](./tscommands.sty) provides the beginner-friendly commands
independently of the template. Place it beside your document (or install it in
your TeX search path), then load it in the preamble:

‘‘‘latex
\documentclass{article}
\usepackage[american]{babel}
\usepackage{tscommands}

\begin{document}
\tsTextBold{Important}
\tsTextItalic{Italic text}
\tsTextQuoteG{German-style quotation marks}
\end{document}
‘‘‘

The package loads its command dependencies. Load Babel with your desired
language before ‘tscommands‘; the quotation commands retain their explicit
styles independently of that language. Commands without input, such as
‘\tsBullet‘, now take no argument. Existing ‘\tsBullet{}‘ calls remain valid:
the empty group is a spacing convention, not a required argument. Commands
with input, such as ‘\tsTextBold{text}‘, retain their arguments.
The package includes text formatting, symbols, captions, citations, footnotes,
images, appendix helpers, the ‘\tsToday{}‘ date format, and ‘tsLTItemize‘.

‘\tsToday{}‘ uses MM/DD/YYYY for ‘american‘ and DD.MM.YYYY for other languages,
with two-digit months and days. It follows ‘\tsDocumentLanguage‘ when defined;
```

otherwise it follows Babel's active language. Its format selection is local and
does not change subsequent ‘\today‘ or ‘\formatdate‘ output.
Babel's ‘english‘ language name is treated as American in standalone usage,
since Babel can report ‘english‘ when loaded with the ‘american‘ alias.

Document metadata, header/footer configuration, TOC leaders, the compatibility
‘P‘ column alias (equivalent to ‘p‘), and numeric minipage footnotes remain in
[TSTemplate-Commands.tex](./TSTemplate-Commands.tex). The package does not enable
appendix mode, build a bibliography, or set the document's layout; these remain
the document author's responsibility.

Dependency and option contract

‘tscommands‘ loads the following direct dependencies:

Dependency	Used for	Fresh-load options
‘babel‘	Explicit quotation symbols and active-language detection	None
‘graphicx‘	Image inclusion	None
‘fancyvrb‘	Verbatim appendix files	None
‘float‘	Exact ‘H‘ figure placement	None
‘datetime‘	Numeric date formats	None
‘ifthen‘	Language and label selection	None
‘ulem‘	Strikethrough text	‘normalem‘
‘enumitem‘	Compact table lists	None
‘overpic‘	Image overlays	‘percent‘
‘nicefrac‘	Fractions	None
‘wasysym‘	Diameter symbol	‘nointegrals‘
‘xcolor‘	Overlay text colors	None
‘contour‘	Outlined overlay text	‘outline‘

Dependencies may load additional packages themselves. All command families
remain available by default; there is no lightweight mode.

Load dependencies with your preferred options **before** ‘tscommands‘. Existing
package configurations take precedence, including for ‘ulem‘, ‘overpic‘,
‘wasysym‘, and ‘contour‘; their fresh-load defaults are not reapplied. For
example:

```latex
\usepackage[ngerman]{babel}
\usepackage[dvipsnames]{xcolor}
\usepackage{wasysym}
\usepackage{tscommands}
```

Alternatively, use ‘\PassOptionsToPackage{options}{package}‘ before
‘tscommands‘ when the dependency has not yet been loaded. For the four
option-bearing dependencies, these passed options are combined with the listed
fresh-load defaults; preload the package when you want to avoid those defaults.
Adding new package options after ‘tscommands‘ has loaded that dependency can
still cause LaTeX's normal option-clash error.

Respecting a preloaded package also respects its behavior: ‘ulem‘ without
‘normalem‘ may change ‘\emph‘, and a differently configured ‘contour‘ may render
overlays differently. Image helpers explicitly use ‘permil‘ coordinates, so
their overlay position does not rely on ‘overpic‘'s global coordinate default.
German/French quote styles and the explicit ‘\textit‘ formatting command remain
independent of the document language.

Template feature switches

```
### Stable reference identifiers

Caption, image, and appendix helpers accept an optional identifier before their
existing arguments:

‘‘‘latex
\tsCaptionLabelFigure[logo-example]{A formatted or repeated caption}
\tsCaptionLabelTable[results]{Results}
\tsCaptionLabelFormula[energy]{E = mc^2}
\tsImage[logo-example]{logo.png}{Company logo}{width=0.5\linewidth}
\tsAppendixSection[notes]{Additional notes}
\tsAppendixIncludeFile[readme]{README.md}
‘‘‘

Use ‘\ref{fig:logo-example}‘, ‘\ref{tab:results}‘, ‘\ref{for:energy}‘,
or ‘\ref{appendix:readme}‘ to reference them. IDs omit the prefix and should be
unique within that prefix; simple letters, numbers, and hyphens are recommended.
All four image helpers (‘\tsImage‘, ‘\tsImageF‘, ‘\tsImageO‘, ‘\tsImageOF‘)
support this leading optional ID.

Omitting the ID, or supplying ‘[]‘, preserves the old caption/filename-based
label for caption, image, and appendix helpers. Prefer explicit IDs for new
content: changing a caption then leaves references stable, and repeated
captions can have different IDs.

### Image placement

All four image helpers accept a second optional argument for figure placement:

‘‘‘latex
\tsImage[logo][htbp]{logo.png}{Company logo}{width=0.5\linewidth}
\tsImage[][htbp]{logo.png}{Company logo}{width=0.5\linewidth}
‘‘‘

The first optional argument is always the ID; use ‘[]‘ when specifying placement
without an ID. Omitting placement retains ‘H‘ (exactly here), including existing
calls with only an ID or neither optional argument. ‘htbp‘ allows LaTeX to place
the figure here, at the top or bottom of a page, or on a float page. ‘!‘ can relax
some float-layout restrictions (for example ‘!htbp‘); it does not mean "exactly
here". Figures remain indivisible and must fit on a page.

### Numbered formulas

Despite its historical name, ‘\tsCaptionLabelFormula[id]{formula}‘ now creates
an ‘equation‘, not a figure caption. Its argument is math source without ‘$‘
delimiters. Call it in document text, not inside another math environment or
as a caption in a figure/table. It increments the equation counter, not the
figure counter, and ‘\ref{for:id}‘ returns its equation number.

Without an ID (or with ‘[]‘), it creates a label from the formatted equation
number, for example ‘for:1‘. Prefer an explicit ID to keep references stable
when equations are inserted or reordered. This intentionally replaces the
former caption-text argument and caption-based numbering. For a captioned
formula illustration in a figure, use ‘\tsCaptionLabelFigure‘ instead.

### Enabling and disabling features

In ‘TSTemplate-Meta.tex‘, use the corresponding ‘...false‘ command to disable
a feature (for example, ‘\UseTsGlossaryfalse‘).

- Glossary: disables package loading, entry definitions, and glossary output.
  Calls to ‘\gls‘ in your content must be guarded by ‘\ifUseTsGlossary‘.
```

```
- Citations: disables bibliography output. The reusable `\tsCitation` command
  remains available, but live citations must be guarded by `\ifUseTsCitation`
  when this document is not producing a bibliography.
- Appendices: skips the appendix file. References to its labels must be guarded
  by `\ifUseTsAppendix`.
- Index: disables index creation and output. Guard index entries with
  `\ifUseTsIndex`.
- TOC, list of figures, and list of tables: control printing of those lists,
  not the availability of sections, figures, or tables.

The supplied examples follow these rules. Their verbatim teaching code remains
visible when a feature is off; disabled live glossary/citation examples are
identified in the text. No supporting commands are silently replaced with
no-ops.

## License

```code
--------------------------------------------------------------------------------
MIT License

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DEALINGS IN THE SOFTWARE.
--------------------------------------------------------------------------------
```

[pdf]: ./TSTemplate.pdf
```

B vscode-explorer-filenesting.json

```
"explorer.fileNesting.enabled": true,
"explorer.fileNesting.patterns": {
    "*.tex": "${capture}*.bib, ${capture}*.pdf, ${capture}*.png, ${capture}*.tex",
    "*.ts": "${capture}.js",
    "*.js": "${capture}.js.map, ${capture}.min.js, ${capture}.d.ts",
    "*.jsx": "${capture}.js",
    "*.tsx": "${capture}.ts",
    "tsconfig.json": "tsconfig.*.json",
    "package.json": "package-lock.json, yarn.lock"
},
```

C vscode-latex-workshop.json

```
// --- Latex workshop settings
"latex-workshop.bibtex-format.sort.enabled": true,
"latex-workshop.bibtex-format.tab": "3 spaces",
"latex-workshop.formatting.latex": "latexindent",
"latex-workshop.latex.autoBuild.run": "never",
"latex-workshop.latex.autoClean.run": "onSucceeded",
"latex-workshop.latex.clean.fileTypes": [
    "*.acn",
    "*.acr",
    "*.alg",
    "*.aux",
    "*.bbl",
    "*.blg",
    "*.fdb_latexmk",
    "*.fls",
    "*.glg",
    "*.glo",
    "*.gls",
    "*.idx",
    "*.ilg",
    "*.ind",
    "*.ist",
    "*.lof",
    "*.log",
    "*.lot",
    "*.nav",
    "*.out",
    "*.snm",
    "*.synctex(busy)",
    "*.synctex.gz",
    "*.synctex.gz(busy)",
    "*.toc",
    "*.vrb",
],
"latex-workshop.latex.clean.method": "glob",
"latex-workshop.latex.recipes": [
    {
        "name": "latexmk (pdfLaTeX)",
        "tools": ["latexmk"],
    },
    {
        "name": "latexmk (LuaLaTeX)",
        "tools": ["lualatexmk"],
    },
    {
        "name": "latexmk (XeLaTeX)",
        "tools": ["xelatexmk"],
    },
    {
        "name": "pdflatex*2",
        "tools": [
            "pdflatex",
            "pdflatex"
        ],
    },
    {
        "name": "pdflatex -> bibtex -> pdflatex*2",
        "tools": [
            "pdflatex",
```

```
            "bibtex",
            "pdflatex",
            "pdflatex"
        ],
    },
    {
        "name": "pdflatex, bibtex, makeglossaries, pdflatex*2",
        "tools": [
            "pdflatex",
            "bibtex",
            "makeglossaries",
            "pdflatex",
            "pdflatex"
        ],
    },
    {
        "name": "pdflatex, makeglossaries, pdflatex*2",
        "tools": [
            "pdflatex",
            "makeglossaries",
            "pdflatex",
            "pdflatex"
        ],
    },
    {
        "name": "lualatex*2",
        "tools": [
            "lualatex",
            "lualatex"
        ],
    },
    {
        "name": "lualatex -> bibtex -> lualatex*2",
        "tools": [
            "lualatex",
            "bibtex",
            "lualatex",
            "lualatex"
        ],
    },
    {
        "name": "lualatex, bibtex, makeglossaries, lualatex*2",
        "tools": [
            "lualatex",
            "bibtex",
            "makeglossaries",
            "lualatex",
            "lualatex"
        ],
    },
    {
        "name": "lualatex, makeglossaries, lualatex*2",
        "tools": [
            "lualatex",
            "makeglossaries",
            "lualatex",
            "lualatex"
        ],
    },
],
"latex-workshop.latex.tools": [
    {
```

```
        "name": "xelatexmk",
        "command": "latexmk",
        "args": [
            "-synctex=1",
            "-interaction=nonstopmode",
            "-file-line-error",
            "-xelatex",
            "-outdir=%OUTDIR%",
            "%DOC%",
        ],
        "env": {},
    },
    {
        "name": "latexmk",
        "command": "latexmk",
        "args": [
            "-synctex=1",
            "-interaction=nonstopmode",
            "-file-line-error",
            "-pdf",
            "-outdir=%OUTDIR%",
            "%DOC%",
        ],
        "env": {},
    },
    {
        "name": "lualatexmk",
        "command": "latexmk",
        "args": [
            "-synctex=1",
            "-interaction=nonstopmode",
            "-file-line-error",
            "-lualatex",
            "-outdir=%OUTDIR%",
            "%DOC%",
        ],
        "env": {},
    },
    {
        "name": "latexmk_rconly",
        "command": "latexmk",
        "args": [
            "%DOC%"
        ],
        "env": {},
    },
    {
        "name": "pdflatex",
        "command": "pdflatex",
        "args": [
            "-synctex=1",
            "-interaction=nonstopmode",
            "-file-line-error",
            "%DOC%",
        ],
        "env": {},
    },
    {
        "name": "bibtex",
        "command": "bibtex",
        "args": [
            "%DOCFILE%"
```

```
        ],
        "env": {},
    },
    {
        "name": "rnw2tex",
        "command": "Rscript",
        "args": [
            "-e",
            "knitr::opts_knit$set(concordance = TRUE); knitr::knit('%DOCFILE_EXT%')",
        ],
        "env": {},
    },
    {
        "name": "jnw2tex",
        "command": "julia",
        "args": [
            "-e",
            "using Weave; weave(\"%DOC_EXT%\", doctype=\"tex\")"
        ],
        "env": {},
    },
    {
        "name": "jnw2texmintex",
        "command": "julia",
        "args": [
            "-e",
            "using Weave; weave(\"%DOC_EXT%\", doctype=\"texminted\")",
        ],
        "env": {},
    },
    {
        "name": "tectonic",
        "command": "tectonic",
        "args": [
            "--synctex",
            "--keep-logs",
            "%DOC%.tex"
        ],
        "env": {},
    },
    {
        "name": "makeglossaries",
        "command": "makeglossaries",
        "args": [
            "%DOCFILE%"
        ],
    },
    {
        "name": "lualatex",
        "command": "lualatex",
        "args": [
            "-synctex=1",
            "-interaction=nonstopmode",
            "-file-line-error",
            "-outdir=%OUTDIR%",
            "%DOC%",
        ],
    },
],
"latex-workshop.linting.chktex.enabled": true,
"latex-workshop.view.pdf.viewer": "browser",
```

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[1] *Latex Font Catalogue*. https://tug.org/FontCatalogue/. Online checked on 16.06.2022.

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